A Comparative Study of Wav2Vec 2.0 and Compact CNN Models for Cross-Corpus Speech Dysfluency Detection

Abstract. We compare two model families for cross-corpus speech dysfluency detection: self-supervised Wav2Vec 2.0 and compact mel-spectrogram CNNs. F1 depends on the decision threshold as well as the model, and thresholds calibrated on one corpus need not transfer. Seven configurations are compared: five independent binary Wav2Vec2 classifiers (arm01); two shared-backbone multitask variants frozen for 3 and 20 epochs (arm02, arm03); and four compact mel-spectrogram CNNs with different sequence aggregators (arm04–arm07). All are trained on 29,296 clips from SEP-28K and UCLASS, then evaluated under a clip-level in-distribution split that shares speakers and a speaker-independent cross-corpus split on 53 English Project Boli recordings. In-distribution, arm01 attains the highest macro F1 (0.570, 95% CI 0.554–0.587, validation-tuned thresholds), with arm02 close behind (0.556, 0.540–0.572) at $\approx$4.9$\times$ fewer parameters; the intervals overlap. Cross-corpus, the ordering inverts: an all-positive classifier attains macro F1 0.550 (0.490–0.604), which nothing exceeds, because the CNNs reach their F1 by labelling nearly every external clip positive (macro predicted-positive rate 0.879, 95% CI 0.849–0.909, against a base rate of 0.408) while the Wav2Vec2 configurations collapse toward all-negative (0.079–0.102). Threshold-free metrics reverse the ranking again: the task-adapted Wav2Vec2 configurations rank external clips best (macro AUROC 0.620–0.621, AUPRC 0.546–0.557) and the CNNs lie near chance. The predicted-positive rate shows the CNN operating points already sat far above the in-distribution base rate, so the CNN advantage under thresholded F1 is consistent with operating-point drift and does not imply superior ranking. Thresholded F1 alone therefore cannot support cross-corpus claims. Each configuration was trained once; intervals reflect clip-level resampling, and the external analysis is exploratory.

Keywords: Speech dysfluency detection · Wav2Vec 2.0 · spectrogram CNNs · cross-corpus evaluation · distribution shift · speaker-independent evaluation.

1 Introduction

Stuttering affects approximately 1% of the global adult population and up to 5% of children [16]. It is characterized by involuntary repetitions, abnormal prolongations, and pauses (blocks), and it affects psychosocial well-being. Clinical assessment relies on subjective listening and waveform inspection, which

is time-consuming, subject to inter-rater variability, and does not scale; this has motivated substantial work on automatic stuttering identification [14]. Self-supervised speech representation learning, and Wav2Vec 2.0 in particular [2], has since produced strong downstream results from raw audio. Applied to dysfluency detection, however, this work is dominated by a single evaluation style: models are trained and tested on a common corpus, and performance is summarized with per-class and macro F1 at a threshold tuned on held-out data from that same corpus. Such F1 measures two things at once: the quality of the model's ranking of clips, and the suitability of a threshold calibrated on data resembling the test data. Only the first should transfer to new speakers and corpora. The two can move in opposite directions under shift: a threshold that maximizes validation F1 in-distribution may fall in a degenerate region of the ROC curve once the score distribution changes, leaving F1 dominated by class base rates rather than discrimination. The available external corpora are small enough that a single operating point can dominate the reported result.

Cross-corpus and speaker-independent evaluation is not new in principle. Bayerl et al. [5] studied cross-corpus stuttering detection as a multi-label problem, and [6] showed that the choice of dataset partitioning materially changes reported results. The interaction between the two metric families under shift is less well characterized: how much of a cross-corpus F1 difference is attributable to ranking quality, and how much to operating-point drift. Without that separation, comparisons between representation families risk being read as statements about generalization when they concern threshold placement. We address this with a controlled comparison of seven configurations, Wav2Vec 2.0 models and compact spectrogram CNNs, under matched protocols, reporting thresholded and threshold-free views side by side. Our research questions are as follows.

- **RQ1**: Under clip-level in-distribution evaluation, how do Wav2Vec2-based and compact spectrogram CNNs compare in macro and per-class F1?
- **RQ2**: On a speaker-independent external corpus, how do their measured F1 scores and their ordering change?
- **RQ3**: How much of the cross-corpus F1 behaviour is explained by operating-point drift (a mismatch between predicted-positive rate and base rate) rather than by a change in ranking quality?
- **RQ4**: Do threshold-free metrics (AUROC, AUPRC) give a materially different account than thresholded F1?

No configuration generalizes across corpora reliably (Section 5). Our contributions are: (i) a matched comparison of Wav2Vec 2.0 and compact spectrogram CNNs under in-distribution and speaker-independent cross-corpus protocols; (ii) an explicit separation of ranking quality from operating-point placement, using the predicted-positive rate measured on both splits under identical thresholds, which shows the two metric families yielding opposite orderings externally; (iii) the observation that no configuration exceeds an all-positive baseline on external F1, a negative result; and (iv) a per-class analysis identifying interjection as easiest and block as hardest.

2 Related Work

2.1 Detection Approaches

Early systems used handcrafted features (jitter, shimmer, spectral flux, formant trajectories) with SVMs, HMMs, or GMMs [14] and generalized poorly across speakers, accents, and conditions. End-to-end deep learning removed the manual feature-engineering step: CNNs on mel-spectrograms with recurrent layers showed early promise for multi-type dysfluency detection [12], as did RNNs and time-delay networks for temporal dependencies [13]. These remain attractive where labelled data and compute are scarce, and define the compact end of the capacity range compared against here. Self-supervised pre-training has since become dominant: Wav2Vec 2.0 contrasts quantized latent speech units against masked waveform spans and generalizes across languages when pre-trained multilingually (XLSR-53 [9]), HuBERT [11] predicts masked hidden units from offline clustering, XLS-R [1] scales the objective to 128 languages, and WavLM [8] adds denoising objectives. Bayerl et al. [4] applied Wav2Vec2 to multi-task stuttering detection on FluencyBank (macro F1 0.56–0.63 across five types) and to UCLASS therapy sessions [7]; on the standardized SEP-28k-E partition per-class F1 varies widely by type (interjection 0.78, prolongation 0.53, block 0.30) [15]. These models encode linguistic content together with paralinguistic attributes such as speaker identity and recording condition; how much corpus-specific information a representation carries is the trade-off our comparison probes.

2.2 Corpora and Metrics Under Shift

The three corpora used here (Section 3) differ in speakers, recording conditions, label definitions, and annotation granularity, so generalization to unseen conditions is difficult to assess. Language is not among the differences our evaluation exercises, because the ingested Boli portion is English-only (Section 3). Bayerl et al. [5] framed cross-corpus detection as a multi-label problem, motivated by dysfluent events co-occurring rather than appearing in isolation, and showed that partitioning changes reported results substantially [6]; most of this literature still reports F1 at a threshold chosen on in-distribution data, the practice whose adequacy we examine. Our study is methodological: we hold the model set fixed, vary the evaluation protocol, and report thresholded and threshold-free metrics for the same predictions.

F1 is the harmonic mean of precision and recall at a chosen operating point, so its value depends on the threshold as well as on the score ranking. AUROC summarizes ranking over all thresholds but can look optimistic when positives are rare, because a large negative class inflates the area; AUPRC weights precision by the positive class, is the more informative of the two under imbalance, and has the positive-class prevalence rather than 0.5 as its chance level. Predictive calibration is a separate property that we do not measure [10]. We report thresholded F1, threshold-free AUROC and AUPRC, the AUPRC lift over the prevalence prior, and the predicted-positive rate (PPR): at a fixed threshold, a PPR far from

the base rate indicates displaced operating-point placement rather than lost discriminative ability. Because the same thresholds are applied to both evaluation sets, the difference in PPR between them measures drift directly.

3 Corpora and Models

The system is a clip-level classification pipeline that predicts which of the five dysfluency types are present in a 3-s audio window. All models share one preprocessing module and are exposed through a common registry API.

3.1 Source Corpora and Clip Construction

SEP-28K contributes 32,321 clips (28,177 podcast clips and a 4,144-clip reannotation of adult FluencyBank speech); UCLASS contributes 4,712 therapy-session clips in SEP-28K format; and Project Boli contributes 54 clips with word-level annotations and timestamps. Boli is confined to the test split, so no external speaker is seen in training. The three are merged into 37,087 manifest rows, with per-clip interval CSVs recording events as `(start_sec, end_sec, type)` tuples; clips whose audio is missing or empty are dropped, leaving 36,674 usable clips. The 36,620 non-Boli clips are partitioned 80/10/10 at random under seed 42 into 29,296 training, 3,662 validation, and 3,662 test clips, and all 54 Boli clips are pinned to the test split, giving a test partition of 3,716 rows. Audio is resampled to 16 kHz mono, DC-removed, peak-normalized, and silence-trimmed.

Evaluation enumerates the audio files in a split directory and keeps those with a matching label CSV and a non-empty body, so the scored unit is the distinct recording rather than the manifest row; on the test partition this yields 3,715 scored clips, of which 53 are Boli, and the same rule gives the Boli count of 53 used throughout.

Construction of the Project Boli subset The external subset is derived by a four-step procedure from the released Boli corpus. First, all `.txt` files in the `Transcripts` directory are enumerated in sorted order; each is a word-level annotation list whose lines carry a start time, an end time, and a dysfluency code. Second, each transcript is paired with audio by name (direct filename lookup, then a mapped lookup through a fixed task table, then a recursive basename search); transcripts with no resolvable audio are discarded. Third, each transcript is parsed line by line, lines with fewer than three fields are skipped, and the remaining codes (*B*, *PR*, *SR*, *WR*, *IN*) are mapped to block, prolongation, sound repetition, word repetition, and interjection; transcripts yielding no recognized code are discarded. The 54 survivors each contribute one manifest row whose five binary labels are set by whether any annotated word carries the code. Fourth, all 54 rows are pinned to the test split and never seen in training; scoring enumerates distinct audio filenames, so they resolve to 53 scored recordings, and all external results use those 53 clips. Because the pairing key is the audio filename, the

scored unit is the recording rather than the annotation row; the reports retain only counts, so we cannot say which filter removed the single row.

Three properties of the released corpus matter for interpretation. First, each annotated file is built around a single stuttered word [3], so the external set is enriched for dysfluent content by construction and is not a sample of naturalistic speech. Second, the released corpus is multilingual (28 participants: 22 Hindi, 2 Telugu, 2 Bengali, 1 Marathi, and 1 Assamese mother tongue; roughly 2.5 h), but our ingestion path resolves only the English task codes, so all 54 ingested clips are English. The protocol therefore changes speakers, recording pipeline, and annotation convention, but not language: it is a cross-corpus evaluation and not a cross-lingual one, and cross-lingual robustness remains untested. Third, the retained audio is the first 3 s of each file (Section 3.3), whereas the Boli label derives from the whole file, so a recording longer than 3 s may carry a label not audible in the segment the model receives.

3.2 Annotation Harmonization

The three corpora use different conventions, which we map onto five common binary clip-level labels: prolongation, block, sound repetition, word repetition, and interjection. SEP-28K labels are adopted as-is, with a clip positive for a type when at least two annotators marked it (`SEP28K_MIN_VOTES` $= 2$); requiring two votes is a substantive choice, since accepting any single vote inflates positives several-fold. Because each UCLASS clip is a 3-s window centered on a stutter event, each present label is materialized as a weak interval spanning 1.25–1.75 s. Boli tokens (*B*, *PR*, *SR*, *WR*, *IN*) map through a fixed code table to the same five classes (Section 3.1), a clip being positive when any annotated word carries the class. All models therefore consume identical five-dimensional binary targets.

3.3 Preprocessing and the 3-Second Window

SEP-28K clips are natively 3 s, so a 3-s maximum window matches the predominant corpus; because the task is clip-level, an event crossing the window boundary does not alter the target. Longer audio is truncated to its first 3 s and shorter audio is zero-padded, so every configuration receives an identically shaped input. Two consequences recur: silence trimming risks suppressing the pause that defines a block, and for Boli recordings longer than 3 s the label derives from the whole file while the retained segment is its first 3 s, so a positive label need not be present in the audio the model receives (Section 6.2).

3.4 Model Configurations

Seven configurations are evaluated. Trainable parameter counts are 472,845,450 for arm01 (five independent Wav2Vec2 binary classifiers), 97,332,362 each for arm02 and arm03 (Wav2Vec2 multitask, backbone frozen for 3 and 20 epochs),

and 341,322, 274,242, 456,906, and 539,594 for arm04–arm07 (CNN with pooling, single-head CNN with pooling, CNN with LSTM, and CNN with transformer). The CNNs therefore hold 0.3–0.6% of the parameters of arm01. The evaluation artifacts report slightly larger CNN counts because they include 708 non-trainable BatchNorm buffer elements; we quote trainable counts here. Inference latency, throughput, and peak GPU memory were not measured and are not inferred from these counts (Section 5.5).

Model descriptions. arm01 uses five independent binary classifiers on `facebook/wav2vec2-base` (94,569,090 parameters each): raw audio → encoder → `Linear`(768→256) on the frame-level hidden states → mean pooling over time → `Linear`(256→2) → softmax. The projection precedes the pooling, the head has no dropout, and no attention mask is passed, so the temporal reduction is an unmasked mean; fixed 3 s inputs give every clip the same number of frames. Each classifier is trained independently with focal loss ($\gamma = 2.0$), AdamW (lr $= 3 \times 10^{-5}$, weight decay $= 0.01$), a 500-step linear warmup, and mixed precision, with the backbone frozen for 3 epochs and then trained at one tenth of the head rate (3×10^{-6}). arm02 and arm03 feed five parallel per-class heads from one shared encoder, each an MLP on a temporal mean pool (`Linear`(768→768) → Tanh → `Linear`(768→2)), minimizing the summed focal loss: 97.3M parameters in total, only ≈2.8M more than one independent classifier. arm03 differs only in freezing the backbone for 20 epochs instead of 3, isolating freezing duration.

arm04–arm07 share a spectrogram backbone and differ in the sequence aggregator. Input is a log-mel spectrogram with 128 Mel bands, a 2048-point FFT, and hop length 512. The encoder is four convolutional blocks with 32, 64, 128, and 128 channels, each with a 3×3 kernel (padding 1), batch normalization, ReLU, a $(2, 1)$ max-pool in all but the last block, and 2-D dropout at rate 0.4. The three pools leave a $128 \times 16 \times 94$ tensor (channels × mel bins × frames). This is aggregated by global mean pooling plus `Linear`(128→128) (arm04); a single-layer unidirectional LSTM with hidden size 128 over the frequency-pooled time axis, using the final hidden state (arm06); or a one-layer transformer (dimension 128, four heads, feed-forward width 512, dropout 0.4) followed by temporal mean pooling (arm07). Five per-class two-layer MLP heads (`Linear`(128→128) → Tanh → `Linear`(128→2)) produce the logits. The CNNs are trained from scratch, with no backbone freezing or pre-training, using binary cross-entropy on the positive logit with a per-class positive weight equal to the negative-to-positive ratio of the training pool, summed over classes. The families thus differ in training objective as well as in representation, which we note as a confound. arm05 is the single-head variant of the pool model: the head block is built once per class rather than five times, and one model is trained per dysfluency type (five in total), mirroring the five-binary Wav2Vec2 design at a much smaller scale.

4 Experimental Setup

4.1 Training Protocol

All models are trained on a single NVIDIA GPU with seed = 42, 20 epochs and early stopping (patience = 5), mixed precision, batch size 8 (binary classifiers) or 16 (multitask and CNN), and 3 s audio at 16 kHz. Training reads the training directory only: it derives its own held-out portion, a 20% stratified subsample (5,857 clips, seed 42) used for early stopping and threshold tuning, so parameters are fitted on 23,439 clips and the prepared validation split is never opened. No gradient clipping or input augmentation is applied; an earlier gradient-clipping configuration was abandoned because, on a 94.6M-parameter model with a randomly initialized head, the clip dominated the head's early gradient signal. Each configuration is trained once, so every reported value is a single-run point estimate.

4.2 Threshold Selection

Thresholds are tuned on the same 20% subsample by sweeping 17 fixed values from 0.1 to 0.9 in steps of 0.05 and, per class, selecting the value maximizing F1 rather than Youden's J. Because that subsample is also the early-stopping set, the tuned thresholds are mildly optimistic; we treat this as a limitation rather than as test-set leakage, since the test split is never used for any selection. arm02 uses thresholds of 0.55, 0.35, 0.40, 0.35, and 0.45 for prolongation, block, sound repetition, word repetition, and interjection. No test-set or cross-corpus threshold fitting is performed: the same thresholds are applied unchanged to the external set, the transfer that Sections 5 and 6 examine.

4.3 Reported Metrics

For each configuration, protocol, and class we report precision, recall, F1, AUROC, and AUPRC, plus macro averages (the unweighted mean over the five classes), which keep classes of differing prevalence from being dominated by the most frequent. AUPRC is reported as a lift over its chance level, the positive-class prevalence. Precision and recall are omitted from the main tables, having fixed the threshold in advance, but are recoverable from F1 and the PPR with the confusion counts.

4.4 Evaluation Protocols

In-distribution, clip-level (Test). The test partition holds 3,716 manifest rows, of which 3,662 come from SEP-28K and UCLASS and 54 from Boli. The partition operates on clips, not speakers, so speakers may appear in both training and test for the SEP-28K and UCLASS portions; these numbers are an upper-bound, not a speaker-independent, benchmark. Results are reported over the 3,715 scored clips, of which the 53 scored Boli recordings ($\approx$1.4%) are out-of-domain. The macro base rate is 0.143.

Speaker-independent, cross-corpus (Boli). The 53 scored Boli recordings are unseen during training, with different speakers, recording conditions, and label-construction procedure (Section 3). Per-class support is small: PR 16, BL 35, SR 34, WR 14, IN 9, giving a macro base rate of 0.408. We treat all external conclusions as exploratory.

Two trivial baselines accompany the models: the *all-positive* classifier, which labels every clip positive for every class, and the *random-at-rate* classifier, which labels clips positive with probability equal to the class base rate. All-positive is the more informative under heavy imbalance: it attains recall 1 by construction and its F1 is determined entirely by the base rate, so a model that does not exceed it has not demonstrated usable discrimination; the expected macro F1 of random-at-rate equals the macro base rate.

4.5 Statistical Uncertainty

Uncertainty is quantified by nonparametric bootstrap resampling of clips: for each reported quantity we resample the evaluated clips with replacement 10,000 times, recompute the metric, and take the 2.5th and 97.5th percentiles. Resampling is unstratified, so it propagates variability in class prevalence as well as in the decisions; on the 53-clip external set the former dominates for the trivial baselines. F1 at a fixed threshold and the predicted-positive rate are functions of the confusion counts alone, so their bootstrap operates exactly on the stored confusion matrices. These intervals capture clip-level sampling variability only: each configuration was trained once, so they exclude training seed, and overlapping intervals should not be read as a real difference in quality. AUROC and AUPRC are point estimates without intervals, because both depend on the full per-clip score vector, which was not retained; we therefore treat the threshold-free ordering as directional evidence.

5 Results

5.1 In-Distribution Results (RQ1)

Table 1 reports macro F1 for all seven configurations under fixed (0.5) and tuned thresholds on both evaluation sets, with the two trivial baselines. Parenthesized intervals are 95% bootstrap confidence intervals (Section 4.5).

Under the clip-level protocol, arm01 attains the highest measured test macro F1 at tuned thresholds and arm02 follows closely: 0.570 (0.554–0.587) against 0.556 (0.539–0.572), with substantially overlapping intervals on 3,715 clips. Since the intervals describe clip sampling alone and each configuration was trained once, we report the pair as indistinguishable and do not treat arm01 as the better model. Both exceed every CNN by a wide margin, their intervals lying far above the CNN range. The CNNs reach macro F1 of 0.25–0.27 at tuned thresholds, above the random-at-rate baseline (0.143) but only marginally above the all-positive baseline (0.247); for arm06 the two intervals overlap. Freezing duration

Table 1. Macro F1 (unweighted mean over five classes) with 95% bootstrap intervals on tuned columns. Test: clip-level in-distribution split (3,715 clips, speakers shared with training); Boli: speaker-independent external split (53 recordings). Tuned thresholds are unchanged on Boli; baselines are untuned.

Arm	Test F1 (0.5)	Test F1 (tuned)	Boli F1 (0.5)	Boli F1 (tuned)
arm01_5x_w2v2	0.573	0.570 (0.554–0.587)	0.254	0.292 (0.169–0.391)
arm02_mt_frz3	0.542	0.556 (0.539–0.572)	0.162	0.238 (0.127–0.332)
arm03_mt_frz20	0.141	0.378 (0.362–0.394)	0.046	0.214 (0.119–0.298)
arm04_cnn_pool	0.149	0.256 (0.245–0.266)	0.351	0.332 (0.254–0.398)
arm05_cnn_single	0.212	0.262 (0.251–0.272)	0.450	0.476 (0.389–0.547)
arm06_cnn_lstm	0.242	0.249 (0.240–0.257)	0.451	0.508 (0.440–0.566)
arm07_cnn_tf	0.269	0.271 (0.259–0.283)	0.421	0.345 (0.260–0.419)
All-positive	0.247	0.247 (0.239–0.254)	0.550	0.550 (0.490–0.604)
Random@rate	0.143	0.143	0.408	0.408

is the one architectural factor with a large and unambiguous effect: arm03, which freezes the backbone for 20 epochs rather than 3, falls to 0.378 (0.362–0.394), a relative reduction of ≈32% from arm02 with non-overlapping intervals, evidence that task adaptation of the encoder is necessary here. Its fixed-0.5 score of 0.141 also falls below random-at-rate, suggesting poorly distributed rather than merely poorly ranked scores.

5.2 Cross-Corpus Results at Tuned Thresholds (RQ2)

On the 53-clip external set, the all-positive baseline attains macro F1 0.550 (0.490–0.604), which no configuration exceeds; the best model, arm06, reaches 0.508 (0.440–0.566). The intervals overlap, so this compares point estimates rather than a separated difference, and the CNN ordering is likewise unresolved: the four CNN intervals span 0.254 to 0.566 and overlap extensively. The one clear separation is between the best CNN and the two task-adapted Wav2Vec2 configurations, whose intervals (0.169–0.391 and 0.127–0.332) lie below that of arm06. Table 2 gives the per-class breakdown.

Table 2. Per-class F1 on the external set (53 recordings) at the validation-tuned thresholds, transferred unchanged. All-positive labels every clip positive, giving per-class F1 $2s/(n+s)$ for support s and $n = 53$; no configuration exceeds its macro F1. Macro intervals are 95% bootstrap intervals; per-class intervals are omitted because with support 9–35 they are too wide to inform the comparison.

Arm	PR	BL	SR	WR	IN	Macro
arm01_5x_w2v2	0.400	0.250	0.256	0.125	0.429	0.292 (0.169–0.391)
arm02_mt_frz3	0.200	0.154	0.256	0.182	0.400	0.238 (0.127–0.332)
arm03_mt_frz20	0.200	0.233	0.182	0.182	0.273	0.214 (0.119–0.298)
arm04_cnn_pool	0.308	0.491	0.182	0.387	0.290	0.332 (0.254–0.398)
arm05_cnn_single	0.326	0.765	0.508	0.431	0.348	0.476 (0.389–0.547)
arm06_cnn_lstm	0.492	0.795	0.542	0.418	0.290	0.508 (0.440–0.566)
arm07_cnn_tf	0.316	0.320	0.490	0.267	0.333	0.345 (0.260–0.419)
All-positive	0.464	0.795	0.782	0.418	0.290	0.550 (0.490–0.604)

The CNN rows appear to show that compact spectrogram models "generalize best"; that reading is incorrect, and Table 3 shows why. The CNNs reach their higher F1 on block, word repetition, and interjection by labelling nearly every external clip positive: arm06 applies the positive label to all 53 clips for each of those three classes, so its per-class F1 approaches, and for interjection exactly equals, the value a predictor with no learned discrimination attains at the same support. Its macro PPR is 0.879, more than twice the 0.408 base rate, while the Wav2Vec2 configurations move the opposite way at PPR 0.079–0.102, leaving their F1 limited by near-zero recall. Both behaviours follow from where a fixed threshold lands after the score distribution shifts; neither is evidence about ranking quality. The finding is negative: at thresholds never fitted to the external set, no configuration demonstrates usable dysfluency discrimination, and the CNN advantage under thresholded F1 is consistent with operating-point drift rather than superior ranking.

5.3 Threshold-Free Cross-Corpus Results and Operating-Point Drift (RQ3, RQ4)

Because F1 conflates ranking with threshold placement, we also report threshold-free quantities for the same predictions (Table 3), together with the predicted-positive rate measured on the in-distribution split under the same thresholds. AUPRC lift is AUPRC divided by the macro base rate of 0.408, the AUPRC of a random predictor.

Table 3. Threshold-free macro metrics and predicted-positive rate (PPR) from the same predictions as Table 2. AUROC and AUPRC are point estimates; AUPRC lift is relative to the 0.408 base rate. PPR is the mean fraction labelled positive at the tuned thresholds and carries a 95% bootstrap interval. The PPR columns use identical thresholds on both sets, so their difference measures drift directly. Base-rate rows give the in-distribution (0.143) and external (0.408) priors.

Arm	AUROC	AUPRC	AUPRC lift	PPR (Test)	PPR (Boli)
arm01_5x_w2v2	0.620	0.546	1.34	0.134	0.079 (0.049–0.113)
arm02_mt_frz3	0.621	0.557	1.37	0.159	0.102 (0.068–0.140)
arm03_mt_frz20	0.538	0.443	1.09	0.195	0.204 (0.158–0.253)
arm04_cnn_pool	0.473	0.412	1.01	0.572	0.574 (0.528–0.619)
arm05_cnn_single	0.497	0.445	1.09	0.583	0.615 (0.566–0.664)
arm06_cnn_lstm	0.522	0.419	1.03	0.845	0.879 (0.849–0.909)
arm07_cnn_tf	0.500	0.412	1.01	0.365	0.438 (0.381–0.494)
Base rate (random)	0.500	0.408	1.00	0.143	0.408

The threshold-free view inverts the thresholded ordering once more. arm01 and arm02 rank external clips best, with macro AUROC 0.620–0.621 and macro AUPRC 0.546–0.557, a lift of ≈1.3–1.4× over the prevalence prior; arm03 sits near chance; and the CNNs are at or barely above chance (AUROC 0.473–0.522, AUPRC 0.412–0.445). The advantage is modest: these are point estimates, and on 53 clips with at most 35 positives per class and 9 in the worst case, a macro AUROC of 0.62 against 0.52 is well within the range that resampling could

reduce to nothing. We therefore state the result directionally: the threshold-free metrics place the task-adapted Wav2Vec2 configurations above the CNNs, and no configuration comes near reliable cross-corpus detection.

The same table answers RQ3: both PPR columns use identical thresholds, so the comparison is a measurement rather than an inference. For six of seven configurations the 95% interval for external macro PPR excludes 0.408, the exception being arm07 (0.381–0.494); the Wav2Vec2 configurations sit below (0.079–0.204) and the CNNs above (0.438–0.879). The CNN operating points were already displaced in-distribution: at base rate 0.143 they label 0.365–0.845 of clips positive (2.6–5.9 times prevalence) while reaching only 0.249–0.271 macro F1. Their apparent external F1 gain is thus not produced by the shift: the over-positive operating point that cripples in-distribution F1 becomes advantageous externally only because the Boli base rate is nearly three times higher. In relative terms their operating point is *less* displaced externally (arm06: 5.9 times the base rate on test versus 2.2 times on Boli), the opposite of the drift a generalization failure would produce. The task-adapted Wav2Vec2 configurations move the other way under the same thresholds, from 0.134 and 0.159 in-distribution to 0.079 and 0.102 externally, toward all-negative. Neither behaviour is about ranking quality, because AUROC and AUPRC do not move with the threshold and order the two families the other way round.

5.4 Per-Class Analysis

Table 4 gives per-class metrics for arm02 on the clip-level test split at its tuned thresholds.

Table 4. Per-class metrics for the multitask Wav2Vec2 model (arm02) on the in-distribution test split (3,715 clips) at validation-tuned thresholds 0.55, 0.35, 0.40, 0.35, and 0.45 for prolongation, block, sound repetition, word repetition, and interjection. Base rate is support divided by 3,715, the chance level of AUPRC. F1 carries a 95% bootstrap interval.

Class	F1 (tuned)	AUROC	AUPRC	Support	Base rate
Prolongation	0.457 (0.407–0.503)	0.845	0.491	400	0.108
Block	0.429 (0.397–0.460)	0.768	0.444	553	0.149
Sound Rep.	0.621 (0.584–0.656)	0.878	0.624	494	0.133
Word Rep.	0.499 (0.456–0.541)	0.855	0.550	366	0.099
Interjection	0.773 (0.750–0.795)	0.935	0.848	834	0.225
Macro	**0.556 (0.540–0.572)**	**0.856**	**0.592**	—	**0.143**

Interjection is the easiest class, at F1 0.773 (0.750–0.795) with AUROC 0.935, and block the hardest, at F1 0.429 (0.397–0.460) with AUROC 0.768; the intervals are well separated. Read against their chance levels, the AUPRC values in Table 4 are 2.99–5.59 times the base rate, so the ranking signal is real in-distribution even where thresholded F1 is low. One reading is that interjections are stereotyped fillers whose acoustic form differs from fluent speech, whereas blocks are pauses close to ordinary hesitation that depend on prosodic and contextual information a 3-s window and a mean-pooled representation may not

preserve; the mechanisms are not separable here, and the block result fits the window and trimming artifacts (Section 3) equally well.

For every class AUROC exceeds tuned F1, most markedly for prolongation (0.845 against 0.457): the models order dysfluent clips better than the tuned thresholds convert that ordering into decisions. Whether this reflects miscalibration or a mismatch between F1-optimal and accuracy-optimal operating points cannot be settled here, because we do not measure calibration. The arm01 block classifier produces the most false negatives (343) and the most false positives relative to its support (284).

5.5 Computational Cost and Relation to Published Results

Parameter count, storage size, computational cost, and runtime cost are distinct: the first two follow from the weights and a stated precision, the last must be measured. We therefore report parameter counts and their fp32 storage in Section 3 but not latency, throughput, or peak GPU memory; parameter counts give fp32 storage of 1.89 GB for arm01 against 1.1–2.2 MB for the CNNs. Within the Wav2Vec2 family we would select the multitask configuration operationally: 97.3M parameters against 472.8M because the encoder is shared, and one forward pass per clip instead of five. It does not measurably beat arm01. Our best in-distribution macro F1, 0.570 (five-binary) and 0.556 (multitask), falls within the 0.56–0.63 reported for Wav2Vec2 on FluencyBank [4], and our per-class ordering matches the 0.30 (block) to 0.78 (interjection) span on SEP-28k-E [15]; this is an ordering comparison, not a numerical one, since our split shares speakers and the corpora, label definitions, and threshold strategies differ [6].

6 Discussion

6.1 What the In-Distribution Results Show

The Wav2Vec2 configurations capture in-distribution signal that the compact CNNs, at roughly 0.3–0.6% of the parameter count, do not. The most likely explanation is representational capacity, but capacity, pretraining, and objective are not separable in our design, since the families also differ in loss and we vary the architecture as a whole; we therefore do not attribute the gap to a single factor. The one isolable factor, freezing duration, acts in the opposite direction and is unambiguous: withholding task adaptation from the same encoder costs $\approx$32% of tuned macro F1, an effect larger than any difference among the CNNs.

6.2 Why the External Results Are Hard to Interpret

Three features complicate readings of Tables 2 and 3: 53 recordings, per-class support as low as 9, and labels constructed unlike the training corpora. The intervals quantify the first two; the third is most consequential. The strongest model's external macro F1, 0.508 (0.440–0.566), overlaps the all-positive baseline's 0.550 (0.490–0.604), so the negative result concerns point estimates on a

set too small to separate them. The comparison the data support cleanly is the operating-point one: PPR intervals exclude the prevailing base rate for six of seven configurations, and the same thresholds in-distribution show the CNN operating points were already displaced. That measurement, not the F1 ranking, is the substantive content.

Because Boli files are built around an annotated stutter event [3], a positive label reflects a curated event rather than sampling continuous speech and applying a majority vote, as with SEP-28K and UCLASS. The all-positive baseline consequently scores 0.550 here against 0.247 on test, a gap at least partly due to clip selection rather than any model. A second complication is the label–window mismatch, which depresses recall unrelated to representation family; we cannot determine how often it occurs because durations were not retained.

6.3 Proposed Explanations

Two explanations the present experiments do not distinguish are stated as proposals rather than findings.

Operating-point drift. The score distributions of all seven configurations are displaced, so thresholds fixed in-distribution land in different ROC regions and F1 tracks the predicted-positive rate and base rate rather than discrimination. Direct evidence is the paired PPR columns of Table 3, where the families sit on opposite sides of the base rate and the ordering flips when the threshold is removed; because both use the same thresholds, displacement is measured rather than assumed. Why it occurs would require the score distributions themselves; the F1 versus AUROC/AUPRC disagreement is its expected signature.

Corpus-specificity. Higher-capacity Wav2Vec2 representations may encode corpus-specific characteristics (speakers, accents, conditions, annotation conventions) that transfer weakly, while coarse spectral patterns transfer more evenly. The threshold-free ordering is consistent, but so are the label–window mismatch and label provenance; no measurement separates them. A probe classifier on frozen embeddings would be the direct test; we have not run one.

6.4 Threats to Validity

Statistical power and single-run training. This is the most consequential threat, with two sources. Each configuration was trained once from seed 42, so the arm01/arm02 difference of under 0.02 macro F1 is not interpretable even with overlapping intervals, and the arm03 degradation carries no training uncertainty: the intervals in Tables 1–4 describe clip sampling only. The external set's 53 recordings with support 9–35 also explain why external F1 comparisons are point estimates and threshold-free metrics have no intervals. We use "better" and "outperforms" only where a difference exceeds plausible seed and sampling variation.

Uncontrolled properties of the external set, and untested explanations. The external labels are built differently from training labels, the clips are English-only and do not test language shift, and truncation to the first 3 s may decouple labels from retained audio. The window and trimming policy is fixed across configurations and protocols, so it does not confound comparisons, but it is an uncontrolled absolute factor and a plausible contributor to block difficulty. Tuned F1 depends on validation-optimal thresholds, and without calibration we cannot distinguish a miscalibrated head from a mismatched operating point [10]; the two accounts remain proposals. Capacity, pretraining, and objective are not separated, latency was not measured, and one external corpus cannot distinguish a general transfer failure from a corpus property.

7 Practical Implications and Conclusion

Model selection for this task cannot rest on in-distribution F1, because that split shares speakers with training and the cross-corpus ordering is dominated by where a fixed threshold lands. Reported results should therefore include a speaker-disjoint or cross-corpus split, threshold-free metrics alongside F1, an all-positive baseline, and the predicted-positive rate at the reported threshold.

On RQ1, Wav2Vec2 dominates in-distribution at macro F1 0.55–0.57 against 0.25–0.27 for the CNNs and 0.247 for an all-positive baseline, with the five-binary and multitask variants indistinguishable and excessive freezing costing ≈32% of tuned macro F1. On RQ2 the cross-corpus F1 ordering inverts, with the CNNs appearing to lead, and on RQ3 that inversion is consistent with operating-point drift rather than generalization: at identical thresholds the CNNs label 37–85% and 44–88% of in-distribution and external clips positive against base rates of 0.143 and 0.408, while the Wav2Vec2 configurations fall to 8–10% externally, and no configuration exceeds the all-positive baseline of 0.550. On RQ4 threshold-free metrics reverse the ordering again, placing the task-adapted Wav2Vec2 configurations above the CNNs at macro AUROC 0.620–0.621 against 0.473–0.522. The conclusions we can draw are narrow: F1 alone cannot support cross-corpus claims for this task, and interjection is easier to detect than block under both protocols. We cannot claim cross-corpus deployment, because no configuration exceeded a trivial baseline on the external set and at least three confounding properties remain (Section 3). The cause of the score drift and the modest threshold-free advantage of Wav2Vec2 remain unresolved (Section 6.4).

References

1. Babu, A., Wang, C., Tjandra, A., Lakhotia, K., Xu, Q., Goyal, N., et al.: XLS-R: Self-supervised cross-lingual speech representation learning at scale. In: Proceedings of Interspeech. pp. 2278–2282 (2022). `https://doi.org/10.21437/Interspeech.2022-143`
2. Baevski, A., Zhou, H., Mohamed, A., Auli, M.: wav2vec 2.0: A framework for self-supervised learning of speech representations. Advances in Neural Information Processing Systems **33**, 12449–12460 (2020)

3. Batra, A., Narang, M., Sharma, N.K., Das, P.K.: Boli: A dataset for understanding stuttering experience and analyzing stuttered speech. arXiv preprint arXiv:2501.15877 (2025). https://doi.org/10.48550/arXiv.2501.15877
4. Bayerl, S., Wagner, D., Lohr, D., Nöth, E., Böcker, S.: Multi-task learning for stuttering detection. In: Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). pp. 6867–6871. IEEE (2022)
5. Bayerl, S.P., Wagner, D., Baumann, I., Hönig, F., Bocklet, T., Nöth, E., Riedhammer, K.: A stutter seldom comes alone: Cross-corpus stuttering detection as a multi-label problem. In: Proceedings of Interspeech (INTERSPEECH). pp. 1538–1542 (2023). https://doi.org/10.21437/Interspeech.2023-2026
6. Bayerl, S.P., Wagner, D., Nöth, E., Bocklet, T., Riedhammer, K.: The influence of dataset-partitioning on dysfluency detection systems. In: Text, Speech, and Dialogue (TSD 2022). pp. 423–436. Lecture Notes in Computer Science, Springer (2022). https://doi.org/10.1007/978-3-031-16270-1_35
7. Bayerl, S.P., Wagner, D., Nöth, E., Riedhammer, K.: Detecting dysfluencies in stuttering therapy using wav2vec 2.0. In: Proceedings of Interspeech (INTERSPEECH). pp. 2868–2872 (2022)
8. Chen, S., Wang, C., Chen, Z., Wu, Y., Liu, S., Li, J., et al.: WavLM: Large-scale self-supervised pre-training for full stack speech processing. IEEE Journal of Selected Topics in Signal Processing (JSTSP) **16**(6), 1505–1518 (2022). https://doi.org/10.1109/JSTSP.2022.3188113
9. Conneau, A., Baevski, A., Collobert, R., Mohamed, A., Auli, M.: Unsupervised cross-lingual representation learning for speech recognition. In: Proceedings of Interspeech. pp. 2426–2430 (2021). https://doi.org/10.21437/Interspeech.2021-329
10. Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q.: On calibration of modern neural networks. In: Proceedings of the International Conference on Machine Learning (ICML). pp. 1321–1330 (2017). https://doi.org/10.48550/arXiv.1706.04599
11. Hsu, W.N., Bolte, B., Tsai, Y.H.H., Lakhotia, K., Salakhutdinov, R., Mohamed, A.: HuBERT: Self-supervised speech representation learning by masked prediction of hidden units. IEEE/ACM Transactions on Audio, Speech, and Language Processing **29**, 3451–3460 (2021). https://doi.org/10.1109/TASLP.2021.3122291
12. Kourkounakis, T., Hajavi, A., Etemad, A.: Detecting multiple speech disfluencies using a deep residual network with bidirectional long short-term memory. In: Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). pp. 6089–6093. IEEE (2020)
13. Sheikh, S.A., Sahidullah, M., Hirsch, F., Ouni, S.: StutterNet: Stuttering detection using time delay neural network. In: 2021 29th European Signal Processing Conference (EUSIPCO). pp. 426–430 (2021)
14. Sheikh, S.A., Sahidullah, M., Hirsch, F., Ouni, S.: Machine learning for stuttering identification: Review, challenges and future directions. Neurocomputing **514**, 385–402 (2022)
15. Sušac, P., Bayerl, S.P., Džapo, H.: Stuttering classification and segmentation with attention-based multiple instance learning. In: Proceedings of Interspeech 2026 (2026), arXiv:2606.20338
16. Yairi, E., Seery, C.H.: Childhood Stuttering: Its Nature, Epidemiology, Etiology, and Treatment. Psychology Press (2009)

Disclosure of Interests. The authors declare no competing interests relevant to this article.